ChE 210 Homework Set #5 Due Friday, 27 February 2026

(1) Murphy 3.21

(2) Murphy 3.29

(3) Murphy 3.41 (To find h_f, the final height of the fluid in the tank, you will need to use Excel Solver to solve the nonlinear equation that arises. On page 676 of Murphy, the solution for this problem is given as 5.1 kg of X. A better value is 5.394 kg of X.)

(4) Phosphoric acid (PA) can be produced by reacting calcium phosphate (CP) with a stoichiometric quantity of sulfuric acid (SA). Calcium sulfate (CS) is also a byproduct of this reaction. Two streams enter the reactor: (**Stream 1**) has a species mass flow rate 10,000 kgCP/hr of CP which is suspended in an unknown flow rate of water (W) and (**Stream 2**) a 95 wt% aqueous solution of SA. There are two exit streams: (**Stream 3**) a 40 wt% aqueous solution of PA and (**Stream 4**) a solid of CS and water with a mole ratio of 2 mole of water for each mole of CS, that is, $CaSO_4 \cdot 2H_2O$, which is commonly call gypsum.

(a) Assuming steady operation, solve for all of the unknown flow rates. Show that the mass flow rate into the process equals the mass flow rate out of the process.

(b) If the operation is started when the tank contains 10,000 kg of 20 wt% phosphoric acid aqueous solution, what is the concentration in the tank after one hour?

(5) A tank containing 1000 L of 95 wt% ethanol solution in water operates at 20°C and steady state with a continuous flow in and out of 100 L/min of 95 wt% ethanol. At t = 0, the input ethanol solution flow is suddenly stop and immediately replaced by a flow of 100 L/min of pure water entering the tank. The pure water flowing in is now diluting the ethanol solution in the tank. Assume the total mass of the solution in the tank remains constant. Assume that the tank is well stirred throughout the process.

DATA: Use the webpage:
https://www.handymath.com/cgi-bin/ethanolwater3.cgi?submit=Entry
for the density of an ethanol / water mixture in kg/L at 20℃. This data is also posted on Blackboard under documents.

(a) How long in minutes will it take for the mass fraction of ethanol in the tank to fall from 95 wt% to 5 wt%.

(b) Recalculate the time for the case when the total volume of the solution in the tank remains constant rather than the total mass. Again dilute the ethanol from 95 wt% to 5 wt%. Give the dilution time in minutes.

Hint: Write mass balances on the total mass and the ethanol mass. Decide what the two unknown variables are. The data gives you the mixture density as a function of mass fraction or vice versa. Combine the two mass balances to eliminate one of the unknowns to create one differential equation with one unknown. You will want to simplify this equation and then solve it.

1.

cabin is 600 ft³/min, for ... flushed out? Assume that the flushing operation is conducted so that the air in the cabin is well mixed.

P3.21 A 12-oz. mug of coffee contains 200 mg of caffeine. Caffeine is eliminated from the body at a rate of $dm_{c,sys}/dt = -0.116m_{c,sys}$, where $m_{c,sys}$ is the mass of caffeine in the body, $m_{c,sys}$ is in mg and t is in h. After you chug down a mug of coffee, how long will it take for the caffeine in your body to drop to 100 mg? You drink one 12-oz. mug at 6 A.M. and another at 2 P.M. Plot the caffeine content of your body as a function of time for one 24-h interval. If you are having trouble falling asleep at 11 P.M., would it help much to cut out that second mug?

P3.22 Titanium dioxide (TiO_2) is by far the most widely used pigment in white paint. Specifications for the white pigment ...

$\frac{dm}{dt} = -0.116m$

$m(t) = m_c$

$\frac{dm}{dt} = -km$

$\int \frac{1}{dm} m = \int k\,dt$

$= \ln|m| = -kt + C$

$e \qquad e$

$m = e^{-kt} \cdot e^{c}$

$e^{c} = m_0$

$m(t) = m_0 \cdot e^{-kt}$

$m(t) = 200 \cdot e^{-0.116t}$

$\ln\left|\frac{100}{200}\right| = \frac{200}{200} \cdot e^{-0.116t} \cdot \ln$

$\ln|0.5| = e^{-0.116t} \cdot \cancel{\ln}$

$\ln|0.5| = -0.116t$

$t = \frac{\ln|0.5|}{-0.116} = 5.97$ hours ✓

6am to 2pm = 8hour between

$m(8) = 200 \cdot e^{-0.16(8)}$

$= 79.06$ mg caffine

$200 + 79.06 = 279.06$mg

$200 \cdot e^{-0.116(17)}$

$m(11) = \boxed{28\text{mg}}$

Plot? -5

10/15

2pm to 11pm → 9 hour between

$m(9) = 200 \cdot e^{-0.116(9)} \approx 98.$ mg caffine

2. $\frac{dx}{dt} = 62{,}000 - 57{,}000 - 2700(1 - e^{-0.3t})$

$$\int_0^{x_f} dx = \int_0^8 2300 + 2700e^{-0.3t}\, dt$$

$$x_f = 2300t - \frac{2700}{0.3} e^{-0.3(t)} \Big|_0^8$$

$$= 2300(8) - \frac{2700}{0.3} e^{-0.3(8)} + \frac{2700}{0.3}$$

$$= 26\,583.54$$

molecules after 8

d

Do not invest b/c it did not met the threshold of $x_f < 30{,}000$ after 8 hours

15/15

3. $\frac{dm}{dt} = m_{in} - m_{out}$

$m = \rho \cdot V$

$V = A \cdot h$

$m = \rho \cdot A \cdot h$

$= \rho(\pi r^2)h$

$(1000)\pi(0.5)^2 h$

$= 785.4h$ ✗

$Acc = in - out - \cancel{gen} - \cancel{cons}$

$= 40 - 4\sqrt{h - h_{out}}$

$= 40 - \sqrt{4h - 4h_{out}}$

$= 40 - 2h + 2h_{out}$

$\rho A \frac{dh}{dt} = 40 - 4\sqrt{h-1}$

$\frac{\rho A}{40 - 4\sqrt{h-1}} dh = dt$

$40 - 4\sqrt{h-1} = 4(10 - \sqrt{h-1})$

$\frac{\rho A}{4(10 - \sqrt{h-1})} dh = dt$

$v = \sqrt{h-1}$

$h = v^2 + 1$

$dh = 2v\,dv$

$\frac{\rho A}{4(10-v)}(2v\,dv) = dt$

$\frac{\rho A}{2} \frac{v}{10-v} dv = dt$

$\int dt = \frac{\rho A}{2} \int \frac{v}{10-v} dv$

$\frac{v}{10-v} = -1 + \frac{10}{10-v}$

$\int(-1 + \frac{10}{10-v})dv$

$\int \frac{10}{10-v} dv \rightarrow -10\ln(10-v)$

$w = 10-v$

$dw = -dv$

$t = \frac{\rho A}{2}(-v - 10\ln(10-v)) + C$

$t - t_i = \frac{\rho A}{2}(-v - 10\ln(10-v) - (0 - 10\ln 10))$

$v + 10\ln(10-v) - 22.97394 = 0$

where $v = \sqrt{h(40) - 1}$

Final height? Mass leaked? -10 pts

10/20

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(b) If the operation is started when the tank contains 10,000 kg of 20 wt% phosphoric acid aqueous solution, what is the concentration in the tank after one hour?

$$Ca_3(PO_4)_2 + 3H_2SO_4 + 6H_2O \rightarrow 2H_3PO_4 + 3CaSO_4\cdot 2H_2O$$

1 mol CP → 2 mol PA

1 mol CP → 3 mol gypsum

	MW
$Ca_3(PO_4)_2$	310
H_2SO_4	98
H_3PO_4	98
$CaSO_4\cdot 2H_2O$	172

10,000 kg CP/hr

n_{CP} = 10000 kg CP/hr

$n_{CP} = \frac{10000}{310} = 32.258$ kmol/hr

Sulfuric acid (3 mol per mol CP)

$n_{SA} = 3(32.258) = 96.774$ kmol/hr

convert to mass

$m_{SA} = 96.774(98) = 9484$ kg/hr

stream 2 is 95 wt% SA

Total stream 2 = $\frac{9484}{0.95} = 9983\ \frac{kg}{hr}$

water in stream 2

$0.05(9983) = 499$ kg/hr

$n_{PA} = 2(32.258) = 64.516\ \frac{kmol}{hr}$

$m_{PA} = 64.516(98) = 6323$ kg/hr

$n_{CS} = 3(32.258) = 96.774$ kmol/hr

$m_{CS} = 96.774(172) = 16648$

$n_{H_2O_{rxn}} = 6(32.258) = 143.548\ \frac{kmol}{hr}$

mass water consumed

$143.548(18) = 3484$ kg/hr

40 wt% PA

6323 kg/hr PA

$\frac{6323}{0.4} = 15{,}808$ kg/hr

water stream 3

15808 − 6323 = 9485 kg/hr

water in stream 1

Total water out

12,969 kg/hr

Total water in

W + 499

W + 499 = 12969

W = 12,470 kg/hr

Stream 1

CP = 10,000 kg/hr

W = 12,470 kg/hr

Total = 22,470

Stream 2

9983 kg/hr (95% SA)

Stream 3 (40% PA)

15808 kg/hr

Stream 4

16,648 kg/hr

Total in

22,470 + 9983 = 32,453

Total out

15,808 + 16,618 = 32,456

B) w(t) = mass fraction

$\frac{d(mw)}{dt} = Fw_{in} - Fw$

$m\frac{dw}{dt} = F(w_{in} - w)$

$w(t) = w_{in} + (w(0) - w_{in})e^{-Ft/m}$

↳ 0.3589

$\frac{Ft}{m} = \frac{15808}{10000} = 1.5808$

$e^{-1.5808} = 0.2057$

w(1) = 0.359 → 35.9 wt PA

20/20

5. $\rho_{95} = 0.80424$ kg/L at 95 wt% ethanol

$\rho_5 = 0.98938$ kg/L at 5 wt% enthanol

$\rho_w = 0.9982067$ kg/L for water at 20 C

$V = 1000L$, $q_{in} = q_{out} = 100$ L/min, $W_0 = 0.95$ $w_f = 0.05$

a) $M = \rho_{95} V = (0.80424)(1000) = 804.24$ kg

$\dot{m}_{in} = \rho_w q = (0.9982067)(100) = 99.82067$ kg/min

$\frac{dw}{dt} = -\frac{\dot{m}_{in}}{M} w$ $w(t) = w_0 e^{-(\dot{m}_{in}/M)t}$

$t = \frac{M}{\dot{m}_{in}} \ln\left(\frac{w_0}{w_f}\right)$

$t = \frac{804.24}{99.82067} \ln\left(\frac{0.95}{0.05}\right) = 23.72$ min ✓

b) $\rho = \rho(w)$

$y = w\rho(w)$

$y_0 = w_0 \rho_{95} = 0.95(0.80424) = 0.764028$

$y_f = w_f \rho_5 = 0.05(0.98938) = 0.049469$

$t = \frac{V}{q} \ln\left(\frac{y_0}{y_f}\right) = \frac{1000}{100} \ln\left(\frac{0.764028}{0.049469}\right)$

$= 27.37$ min ✓

30/30